

Device Under Test

We ordered 10 amplifiers from eBay in 2022 for use with Opulent Voice.

From the original listing at <https://www.ebay.com/itm/363233702995>

Microwave RF Power Amplifier Board SBB5089+SHF0589 40MHz-1.2GHz Gain 25DB 10PCS

Specifications:

- Input voltage: 10~30V DC
- Input power: about 5W
- Working frequency: 40MHz~1.2GHz (0.04~1.2GHz)
- Gain: about 25dB (may be higher)
- Power: 2W (may be higher)

Attention:

We measured 80.7% ultra-high efficiency in tests, and the official chip manual also mentioned that there is more than 50% efficiency at P1dB. Overall, this SBB5089+SHF0589 is better than SBB5089+SHF0289.

Introduction

On 24 March 2026, we selected one of the amplifiers at random in order to characterize it for use with the LibreSDR for over the air Opulent Voice testing. We had received a report from Oscar Resto (UPV) that two of these amplifiers burned up when used in their lab.

We connected the input of the amplifier to the output of the DSG821A signal generator. The signal generator was set to 431 MHz. WE connected the output of the amplifier through a 6 dB attenuator to the Rigol RSA5065N Spectrum Analyzer. We fitted a JST-HX power cable to the power connector of the amplifier. We provided 12 volts of power from the DP832 lab power supply.

Signal Generator Results

Input Power	Output Power	Gain	Current Draw
-99 dBm	-71.8 dBm	27.2	0.62 A
-94 dBm	-67 dBm	27.0	0.62 A
-89 dBm	-62 dBm	27.0	0.62 A
-84 dBm	-57 dBm	27.0	0.61 A
-79 dBm	-52 dBm	27.0	0.61 A
-74 dBm	-47 dBm	27.0	0.61 A
-69 dBm	-42 dBm	27.0	0.61 A
-64 dBm	-37 dBm	27.0	0.61 A
-59 dBm	-32 dBm	27.0	0.61 A
-54 dBm	-27 dBm	27.0	0.61 A
-49 dBm	-22 dBm	27.0	0.61 A
-44 dBm	-17 dBm	27.0	0.61 A
-39 dBm	-12 dBm	27.0	0.61 A
-34 dBm	-7 dBm	27.0	0.61 A

-29 dBm	-2 dBm	27.0	0.61 A
-24 dBm	4 dBm	28.0	0.61 A
-19 dBm	8.3 dBm	27.3	0.61 A
-14 dBm	13.6 dBm	27.6	0.61 A
-9 dBm	19 dBm	28.0	0.62 A
-4 dBm	23.6 dBm	27.6	0.63 A
0 dBm	25.3 dBm	25.3	0.60 A
3 dBm	26.0 dBm	23.0	0.56 A
6 dBm	26.5 dBm	20.5	0.52 A
9 dBm	26.7 dBm	17.7	0.50 A

Libre SDR Results

Next test was to remove the signal generator and connect the LibreSDR running Opulent Voice. Measurements would be repeated. However, we noticed something very interesting. The signal level was expected to be about 0 dBm, which would provide enough drive to the amplifier to create enough gain to attempt some over the air tests. When the LibreSDR, running Locutus and Dialogus, was commanded to transmit with PTT and audio frames from Interlocutor, the peak of the main lobe of the MSK signal was at -30 dBm.

The transmit power hardware gain setting was confirmed to be at 0 dBm. This is set through an IIO attribute call. The other possibility was that the I and Q signals were not being generated for transmit at full scale. Investigation turned to the HDL files. A hypothesis of an inadvertent divide by 16 in the `msk_modulator` submodule was developed.

Our code:

```
tx_samples_I <= std_logic_vector(resize(s1s + s2s, SAMPLE_W));
tx_samples_Q <= std_logic_vector(resize(s1c + s2c, SAMPLE_W));
```

Hypothesis:

The `msk_modulator` adds two 12-bit sinusoids (`s1s + s2s`). This produces a 13-bit result. Then `resize(..., SAMPLE_W=16)` sign extends it to 16 bits. This is right-justified in [12:0], with a sign extension in [15:13].

The `axi_ad9361` interprets `dac_data_i0[15:0]` differently by mode. In CMOS (on the Pluto) the radio chip reads [11:0]. Our data is 12 LSBs of sum. And, it works. In LVDS mode (on the LibreSDR) the radio chip reads [15:4]. This is sign extension and the upper 9 bits of the sum. This divides by 16 and may result in -24 dB of attenuation. We need some help figuring out if this is the root cause or not with the substantial unexpected attenuation of the transmitted signal from the LibreSDR.

Additionally, a post from Engineer Zone:

https://ez.analog.com/fpga/f/q-a/112155/axi_ad9361-data-format

This issue was referred to #opulent-voice channel on Slack for further discussion and evaluation.

Questions for Oscar

What was the behavior of the amplifiers from this batch that you received? What happened when they failed? Our amplifiers are running relatively hot, as in the heat sinks heat up, but they are creating gain.

-Abraxas3d

-Sknkwrks